DETERMINISTIC INVENTORY MODELING

Notation

c: unit purchase cost ($/unit)
K: fixed ordering cost ($/order)
i: annual inventory carrying rate ($/$/yr)
h=ic: unit annual inventory holding cost ($/unit/yr)
λ: demand rate (units/yr)
P: production rate (units/yr)
τ: constant lead time (yrs)
r: reorder level (units)

Cycle Length

$$T = \frac{q}{\lambda}$$

Single product: Retail/Procurement setting

$$C(q) = c\lambda + \frac{K\lambda}{q} + \frac{icq}{2}$$

$$q = \sqrt{\frac{2K\lambda}{h}}$$

Single product: Production setting

$$C(q) = c\lambda + \frac{K\lambda}{q} + \left(1 - \frac{\lambda}{P}\right)\frac{icq}{2}$$

$$q = \sqrt{\frac{2K\lambda}{ic\left(1 - \frac{\lambda}{P}\right)}}$$

Reorder level

Given the constant lead time of τ
$r = mod(\lambda\tau, q)$

Discount schemes

A discount scheme is characterized by a set of break points, indexed by $j = 0, \dots, r$, $b_0, b_1, \dots, b_r$, where

1. $b_0 < b_1 < \cdots b_r$ and
2. $b_0 = 0$

along with a set of unit prices $c_0, c_1, \dots, c_r$ where

$$c_0 > c_1 > \cdots > c_r$$

All-units discount scheme

Step 1: Evaluate the economic order quantity value for each price level c_j using

$$q_j = \sqrt{\frac{2K\lambda}{ic_j}} \text{ for } j = 1, \dots, r.$$

Step 2: For each order quantity evaluated in Step 1 and the corresponding lower and upper bounds for the price range, set feasible order quantities using:

1. If $q_j < b_j$ then $q_j^* = b_j$.
2. If $b_j \le q_j < b_{j+1}$ then $q_j^* = q_j$.
3. If $b_{j+1} < q_j$ then there is no candidate solution.

Step 3: Evaluate average annual total cost for each price level c_j using

$$C(q_j^*) = c_j\lambda + \frac{K\lambda}{q_j^*} + \frac{ic_jq_j^*}{2}$$

Step 4: Over all j, select the order quantity with the lowest average annual total cost.

Incremental discount scheme

Step 1: Evaluate V_j values at each price level $c_j$using
$V_0 = 0$
$V_j = c_0(b_1 - b_0) + c_1(b_2 - b_1) + \cdots + c_{j-1}(b_j - b_{j-1})$

Step 2: Evaluate EOQ values at each price level c_j using

$$q_j = \sqrt{\frac{2\lambda(K + V_j - b_jc_j)}{ic_i}} \text{ for } j = 1, \dots, r.$$

Step 3: For each order quantity evaluated in Step 1 and the corresponding lower and upper bounds for the price range, set feasible order quantities using

4. If $q_j < b_j$ then $q_j^* = b_j$.
5. If $b_j \le q_j < b_{j+1}$ then $q_j^* = q_j$.
6. If $b_{j+1} < q_j$ then $q_j^* = b_{j+1}$.

Step 4: Evaluate average annual total cost for each price level c_j and choose the one that yields the minimum average annual total cost using

$$C_j(q_i^*) = \frac{K\lambda}{q_i^*} + \left[V_j + (q_i^* - b_j)c_j\right]\frac{i}{2} + \left[V_j + (q_i^* - b_j)c_j\right]\frac{\lambda}{q_i^*}$$

STOCHASTIC INVENTORY MODELING

Notation

p: unit selling price ($/unit)
c: unit purchase cost ($/unit)
b: unit loss-of-goodwill cost ($/unit)
v: unit salvage revenue ($/unit)
K: fixed ordering cost ($/order)
i: annual inventory carrying rate ($/$/yr)
h=ic: unit annual inventory holding cost ($/unit/yr)
λ: demand rate (units/yr)
q: order quantity (units)
r: reorder level (units)

$n(r)$: expected number of shortages in a cycle when the reorder level is r (units)
s: expected safety stock level (units)
X_L: demand during lead time, a random variable with pdf $f(x_L)$; cdf $F(x_L)$; $E[X_L]$ =; and $Var(X_L) = \sigma_L^2$
μ_D: expected demand
σ_D^2: variance of demand
μ_τ: expected lead time
σ_τ^2: variance of lead time

Leibnitz's Rule

$$\frac{d}{dy}\int_{\ell(y)}^{u(y)} h(x,y)dx = \int_{\ell(y)}^{u(y)} \frac{\partial h(x,y)}{\partial y}dx + h(u(y),y)u'(y) - h(\ell(y),y)\ell'(y)$$

Periodic Review Systems

$$F(q^*) = \frac{c_u}{c_u + c_o}$$

	Underage	Overage
Single-period: Cost min.	c_u	c_o
Single-period: Profit max.	$p - c + b$	$c - v$
Multiple-period: Full back.	b	h
Multiple-period: Lost sales	$p - c + b$	h

Demand during Lead Time (DDLT) Parameters
$\mu_L = \mu_D \mu_\tau$
$\sigma_L^2 = \mu_\tau \sigma_D^2 + \mu_D^2 \sigma_\tau^2$

Continuous Review Systems
$$n(r) = \int_r^\infty (x_L - r)\, f(x_L) dx_L$$
If the demand is normally distributed, we have
$$n(r) = \sigma_L L(z) = \sigma_L L\left(\frac{r - \mu_L}{\sigma_L}\right)$$

Full Backlogging
$s = r - E[X_L]$

$$q^* = \sqrt{\frac{2\lambda[K + bn(r)]}{h}}$$

$$1 - F(r^*) = \frac{qh}{b\lambda}$$

$$r^* = \mu_L + z^* \sigma_L$$

$$C(q,r) = c\lambda + \frac{K\lambda}{q} + h\left(\frac{q}{2} + s\right) + \frac{bn(r)\lambda}{q}$$

Lost Sales
$s = r - E[X_L] + n(r)$

$$q^* = \sqrt{\frac{2\lambda[K + bn(r)]}{h}}$$

$$1 - F(r^*) = \frac{qh}{qh + b\lambda}$$

$$r^* = \mu_L + z^* \sigma_L$$

$$C(q,r) = c\lambda + \frac{K\lambda}{q} + h\left(\frac{q}{2} + s\right) + \frac{bn(r)\lambda}{q}$$

Type I Service Level
$$q = \sqrt{\frac{2K\lambda}{ic}}$$
$F(r) = \alpha$

Type II Service Level
$$q = \sqrt{\frac{2K\lambda}{ic}}$$
$n(r) = (1 - \beta)q$

LOGISTICS

Notation
h: in-transit inventory holding cost ($/unit/year)
λ: demand rate (units/yr)
τ: lead time (years)

In-transit inventory holding cost
$h\lambda\tau$